11 Chemistry — Unit 2 Topic Test 6: Redox

SOLUTIONS AND MARKING GUIDE — teacher copy; release to students only after the test has been sat

Total 40 marks: Section A 10 × 1 mark (separate answer sheet), Section B 30 marks. Writing time 46 min (40 marks × 1.15 min per mark). Students use the VCE Chemistry Data Book (SLC 25 °C and 100 kPa; V_m = 24.8 L mol^{-1} at SLC; R = 8.31 J mol^{-1} K^{-1}; 1 atm = 101.3 kPa; c(water) = 4.18 J g^{-1} $°C^{-1}$; T(K) = T(°C) + 273). **One mark per idea or step; no half marks.** Mark units and significant figures **once per part, at the final answer** (3 significant figures expected unless stated; accept 2–4 where the data allow). **Rounding tolerance:** accept a final answer within ±1 in the last significant figure when it comes from correctly rounded intermediate values; each part states its accepted range. **Consequential marking:** where a later part uses an earlier answer, award full marks for correct method applied to the student's earlier value; parts where this applies are flagged *Consequential*. Energy released or absorbed is a positive magnitude. Equations need correct states.

Section A — Multiple-choice answer key (10 marks)

Key: B C A C A D B D B C. Balance: A ×2, B ×3, C ×3, D ×2; no letter more than 1 in a row. Each wrong letter points to one slip or misconception (right-hand column). Difficulty and demand level for each item are shown in italics after the working.

Q	Key	Working for the key	Distractor diagnosis
1	B	A reducing agent causes reduction by giving electrons away, so it is itself oxidised. *[Easy; L1]*	**A** described an oxidising agent; **C** thinks the reducing agent is reduced; **D** reversed electron transfer and mismatched the process
2	C	Strongest reducing agent = the metal whose ion is lowest in the electrochemical series (most negative E°): Al^{3+}/Al is lowest of the four. *[Easy; L2]*	**A** picked a metal from the middle of the series; **B** read the series in the wrong direction; **D** read the series in the wrong direction / picked the metal near the top
3	A	x + 4(−2) = −2, so x = +6. *[Medium; L2]*	**B** ignored the charge on the ion (set the total to 0); **C** used the wrong sign for the charge on the ion; **D** gave the charge on the ion instead of the oxidation number
4	C	C in CO_2: +4 → 0, so each CO_2 gains 4 electrons (supplied by 2 Mg atoms, 0 → +2 each). *[Medium; L3]*	**A** counted one electron per CO_2 molecule (ignored the oxidation-number change); **B** counted the electrons lost by only one Mg atom; **D** doubled the change for carbon (counted 2 × 4)
5	A	V in VO_2^+: x + 2(−2) = +1, x = +5; V in VO^{2+}: x − 2 = +2, x = +4. An electron is gained, so V is reduced. Charge: +2 on each side. *[Hard; L3]*	**B** read the half-equation in reverse / ignored the side the electron is on; **C** hydrogen stays at +1; H^+ only balances the oxygen as water; **D** did not add the charges: left +1 + 2 − 1 = +2, right +2
6	D	In $Cu_2(OH)_2CO_3$: 2x + 2(−1) + (−2) = 0, so x = +2. Copper metal is 0. *[Medium; L2]*	**A** shared a charge of +2 between the two copper atoms; **B** reversed: thinks copper is reduced; **C** gave the total for both copper atoms
7	B	O in O_2: 0 → −2 (reduced), so O_2 is the oxidising agent; Cu: 0 → +2 (oxidised). *[Easy; L2]*	**A** chose the reducing agent (copper is oxidised); **C** no element in H_2O changes oxidation number; **D** carbon stays at +4 and oxygen stays at −2 in CO_2 and in the carbonate
8	D	2Al + $3Cu^{2+}$ → $2Al^{3+}$ + 3Cu ($6e^-$). n(Al) = 0.540 ÷ 27.0 = 0.0200 mol; n(Cu) = 0.0300 mol; m = 0.0300 × 63.5 = 1.91 g. *[Hard; L3]*	**A** used a 1:1 ratio for Al : Cu (did not balance electrons); **B** inverted the 2:3 ratio; **C** used a 1:3 ratio (3 Cu per Al)
9	B	At the anode the metal is oxidised to ions that go into solution, so it loses mass and releases electrons into the wire towards the cathode. The cathode is the positive electrode, so III is false.	**A** thinks the anode gains mass (confuses it with the cathode, where metal is deposited); **C** thinks the cathode is negative (true for electrolysis, not a galvanic cell); **D** thinks the

		[Medium; L3]	cathode is negative in a galvanic cell
10	C	Only oxygen (the oxidising agent) and water are required for rusting. *[Easy; L1]*	**A** ignores the need for water (iron in dry air does not rust); **B** ignores the need for oxygen, the oxidising agent; **D** nitrogen is unreactive; oxygen is the oxidising agent that is needed

Section B — Worked solutions and marking scheme (30 marks)

Each part shows the full worked solution, then the bold mark allocation (one mark per idea), then *Consequential* and *Reject* notes. Difficulty and demand level are shown in italics.

Question 1 — Cleaning up an aquarium (8 marks)

a) (1 mark)

x + 4(+1) = +1, so x = −3

1 mark −3 (written with the sign before the number). *[Easy; L2]*

Reject: 3−, +3, −4.

b) (1 mark)

x + 2(−2) = −1, so x = +3

1 mark +3. *[Medium; L2]*

Reject: +4 (charge on the ion ignored), 3+.

c) (2 marks)

$NH_4^+(aq) + 2H_2O(l) \rightarrow NO_2^-(aq) + 8H^+(aq) + 6e^-$

Check: N 1 = 1; O 2 = 2; H 4 + 4 = 8; charge +1 = −1 + 8 − 6 = +1.

1 mark N and O balanced with H_2O on the left and H^+ on the right (atoms balanced). **1 mark** $6e^-$ on the right so charge balances, with states. *[Hard; L2]*

Reject: Electrons on the left (reduction). OH^- used (not acidic conditions). $8e^-$ or $4e^-$.

d) (1 mark)

Oxygen, O_2.

1 mark O_2 (oxygen). *[Easy; L1]*

Reject: NH_4^+ (the reducing agent). “The bacteria.” H^+.

e) (2 marks)

(c) × 2: $2NH_4^+ + 4H_2O \rightarrow 2NO_2^- + 16H^+ + 12e^-$

O_2 equation × 3: $3O_2 + 12H^+ + 12e^- \rightarrow 6H_2O$

Add and cancel: $2NH_4^+(aq) + 3O_2(g) \rightarrow 2NO_2^-(aq) + 4H^+(aq) + 2H_2O(l)$

1 mark electrons matched ($12e^-$) and cancelled. **1 mark** correct overall equation with H^+ and H_2O simplified and states. *[Medium; L2]*

Consequential: On the half-equation in part c), if it is balanced for atoms.

Reject: Electrons left in the overall equation. H^+ or H_2O on both sides.

f) (1 mark)

$NO_3^-(aq) + 2H^+(aq) + 2e^- \rightarrow NO_2^-(aq) + H_2O(l)$

1 mark fully correct, including states. *[Medium; L2]*

Reject: Electrons on the right. Unbalanced charge.

The oxygen half-equation is supplied so that part e) tests combining, not recall. Part e) is consequential on c).

Question 2 — Cleaning tarnished silver (9 marks)

a) (1 mark)

$Al(s) \rightarrow Al^{3+}(aq) + 3e^-$

1 mark correct with states. *[Easy; L2]*

Reject: Electrons on the left. Al^{2+}.

b) (2 marks)

Ag_2S half-equation × 3 and Al half-equation × 2 ($6e^-$ each):

$3Ag_2S(s) + 2Al(s) \rightarrow 6Ag(s) + 2Al^{3+}(aq) + 3S^{2-}(aq)$

1 mark electrons balanced ($6e^-$) and cancelled. **1 mark** correct coefficients 3 : 2 : 6 : 2 : 3 with states. *[Medium; L2]*

Consequential: On the half-equation in part a).

Reject: Electrons shown in the overall equation. $Ag_2S + Al \rightarrow 2Ag + Al^{3+} + S^{2-}$ (charge not balanced).

c) (1 mark)

Aluminium, Al(s).

1 mark Al. *[Easy; L1]*

Reject: Ag_2S. Al^{3+}. $NaHCO_3$.

d) (2 marks)

The aluminium is oxidised at the foil, but the Ag_2S that is reduced is on the surface of the cutlery, so the two reactions happen in different places (like the two electrodes of a galvanic cell).

The electrons released by the aluminium must travel through the metal-to-metal contact to the cutlery; the solution carries ions but not electrons, so without contact there is no path for electrons and the Ag_2S is not reduced.

1 mark the oxidation (Al) and reduction (Ag_2S) occur at different places, so electrons must be transferred from the foil to the silver. **1 mark** electrons can only travel through the metal contact (not through the solution), so without contact no reduction occurs. *[Hard; L3]*

Reject: “So the heat can transfer.” “So the aluminium can react with the silver” with no electron pathway. Electrons travelling through the solution.

e) (2 marks)

$M(Ag_2S) = 2(107.9) + 32.1 = 247.9$ g mol^{-1}; $n(Ag_2S) = 0.124 \div 247.9 = 5.00 \times 10^{-4}$ mol

$n(Al) = 2/3 \times 5.00 \times 10^{-4} = 3.33 \times 10^{-4}$ mol; $m(Al) = 3.33 \times 10^{-4} \times 27.0 = 9.00 \times 10^{-3}$ g (9.00 mg)

1 mark $n(Ag_2S) = 5.00 \times 10^{-4}$ mol and $n(Al) = 3.33 \times 10^{-4}$ mol using the 3 : 2 ratio. **1 mark** 9.00×10^{-3} g (accept 8.99–9.01 × 10^{-3} g, 0.00900 g or 9.0 mg; unit required). *[Medium; L2]*

Consequential: On the ratio in the student’s overall equation in part b).

Reject: 1:1 ratio (0.0135 g). Ratio inverted (0.0203 g).

f) (1 mark)

It is an electrolyte: its ions move through the solution between the foil and the cutlery, completing the circuit (like a salt bridge) so that electrons can keep flowing through the metal contact.

1 mark the solution is an electrolyte: its ions move to carry charge between the foil and the cutlery, completing the circuit. *[Medium; L3]*

Reject: “It cleans the silver.” “It carries the electrons.” “It is the oxidising agent.”

Part d) is about the electron path (metal contact); part f) is about the ion path (electrolyte). Credit an answer to d) that also mentions ions, but the marks in d) are for electron transfer through the contact.

Question 3 — A magnesium–nickel cell (8 marks)

a) (1 mark)

The magnesium electrode (the anode): Mg^{2+}/Mg is below Ni^{2+}/Ni in the electrochemical series, so Mg is the stronger reducing agent and is oxidised.

1 mark magnesium (Mg) electrode. *[Medium; L2]*

Reject: Nickel. "The anode" alone without naming the electrode (0).

b) (2 marks)

Anode: $Mg(s) \rightarrow Mg^{2+}(aq) + 2e^-$

Cathode: $Ni^{2+}(aq) + 2e^- \rightarrow Ni(s)$

1 mark anode half-equation with states, labelled anode. **1 mark** cathode half-equation with states, labelled cathode. Two correct but unlabelled (or swapped) half-equations score 1 mark. *[Medium; L2]*

Reject: Na^+ or SO_4^{2-} reacting. Water reduced at the cathode (Ni^{2+} is the strongest oxidising agent present).

c) (1 mark)

Positive (it is the cathode, where reduction occurs).

1 mark positive. *[Easy; L1]*

Reject: Negative.

d) (2 marks)

Towards the magnesium half-cell.

Mg^{2+} ions form there as magnesium is oxidised, so that solution would build up positive charge; SO_4^{2-} ions move in to balance the charge and keep the solution electrically neutral, so the cell can keep operating.

1 mark towards the magnesium (anode) half-cell. **1 mark** because Mg^{2+} ions are produced there, and the anions balance the extra positive charge (keep the solution neutral). *[Medium; L3]*

Reject: Towards the nickel half-cell. "Negative ions are attracted to the positive electrode." "The sulfate ions carry electrons."

e) (2 marks)

Mg is a stronger reducing agent than Ni, so Mg reacts directly with the Ni^{2+} ions in contact with it (Ni metal deposits on the Mg strip).

The electrons are transferred directly from Mg atoms to Ni^{2+} ions at the metal surface, not through an external wire, so no current flows through the voltmeter (the energy is released as heat instead).

1 mark Mg reacts directly with Ni^{2+} (displacement) when they are in contact. **1 mark** electrons transfer directly at the metal surface rather than through the external circuit, so no current / chemical energy becomes heat, not electrical energy. *[Hard; L3]*

Reject: "Nothing would happen, because there is no salt bridge." "Magnesium does not react with nickel ions."

Part d) tests ion movement in the salt bridge by explanation; part e) links displacement to why the half-cells are kept apart. Electron flow is tested in MC 9.

Question 4 — Iron wire in gel (5 marks)

a) (1 mark)

The iron has been oxidised to iron(II) ions: it has started to corrode.

1 mark iron has been oxidised / lost electrons to form Fe^{2+} (accept "the iron has corroded"). *[Medium; L3]*

Reject: "The iron has been reduced." "Rust is blue."

b) (1 mark)

It is a protective coating (barrier) that stops oxygen and water reaching the iron.

1 mark barrier that keeps O_2 and/or water away from the iron. *[Easy; L1]*

Reject: "The lacquer is oxidised instead." "It makes the iron less reactive."

c) (1 mark)

$Mg(s) \rightarrow Mg^{2+}(aq) + 2e^-$

1 mark correct with states. *[Medium; L2]*

Reject: Electrons on the left.

d) (2 marks)

Piece B: Mg^{2+}/Mg is below Fe^{2+}/Fe in the electrochemical series, so magnesium gives up electrons more readily than iron. The magnesium is oxidised, and the electrons it releases flow into the wire, so the iron is not oxidised and no Fe^{2+} forms, even where the iron is bare.

Piece C: paint protects only while it is unbroken. At the scratch, oxygen and water reach the iron, and no more easily oxidised metal is in contact with it, so the iron itself is oxidised to Fe^{2+} there.

1 mark piece B: Mg is lower in the series than Fe (gives up electrons more readily), so Mg is oxidised instead of the iron. **1 mark** piece C: the paint is only a barrier; where it is broken, O_2 and water reach the iron and nothing else is oxidised in its place, so the iron corrodes there. *[Hard; L4]*

Reject: "Magnesium is more reactive" with no reference to the electrochemical series or electron loss (0 for that mark). "The paint reacts at the scratch."

Part d) compares a metal that is oxidised in place of iron with a barrier coating that has been breached; both points must be tied to the observations.

Mark summary

Section	Marks	Parts
A — multiple choice	10	10 × 1 mark; key A2 / B3 / C3 / D2
B Q1 — Cleaning up an aquarium	8	a 1, b 1, c 2, d 1, e 2, f 1
B Q2 — Cleaning tarnished silver	9	a 1, b 2, c 1, d 2, e 2, f 1
B Q3 — A magnesium–nickel cell	8	a 1, b 2, c 1, d 2, e 2
B Q4 — Iron wire in gel	5	a 1, b 1, c 1, d 2
Total	40	40 marks × 1.15 min = 46 min writing time

Difficulty tally (marks)

Target for every topic test: 25% easy (10 marks), 50% medium (20 marks), 25% hard (10 marks).

	Easy	Medium	Hard	Total
Section A (MC)	4	4	2	10
Section B (short answer)	6	16	8	30
Whole test (marks)	10	20	10	40
Share of marks	25%	50%	25%	100%

Difficulty and demand by item

Item	Marks	Difficulty	Demand
A1	1	Easy	L1
A2	1	Easy	L2
A3	1	Medium	L2
A4	1	Medium	L3
A5	1	Hard	L3
A6	1	Medium	L2
A7	1	Easy	L2
A8	1	Hard	L3
A9	1	Medium	L3

A10	1	Easy	L1
B1a	1	Easy	L2
B1b	1	Medium	L2
B1c	2	Hard	L2
B1d	1	Easy	L1
B1e	2	Medium	L2
B1f	1	Medium	L2
B2a	1	Easy	L2
B2b	2	Medium	L2
B2c	1	Easy	L1
B2d	2	Hard	L3
B2e	2	Medium	L2
B2f	1	Medium	L3
B3a	1	Medium	L2
B3b	2	Medium	L2
B3c	1	Easy	L1
B3d	2	Medium	L3
B3e	2	Hard	L3
B4a	1	Medium	L3
B4b	1	Easy	L1
B4c	1	Medium	L2
B4d	2	Hard	L4

Cognitive demand

Demand level	Marks	Share
L1 Recall / identify	6	15%
L2 Routine procedure	20	50%
L3 Interpret / explain	12	30%
L4 Evaluate / error direction	2	5%

Study design coverage

Key knowledge (Unit 2)	Where assessed
Oxidation and reduction; oxidising and reducing agents	A1, A4, A5, A7; B1d, B2c
Oxidation numbers	A3, A6; B1a, B1b
Half-equations and overall equations, including acidic conditions	A5; B1c, B1e, B1f, B2a, B2b, B3b
Reactivity series and metal displacement (electrochemical series)	A2, A8; B3a, B3e, B4d
Corrosion: conditions and prevention (coatings, sacrificial protection)	A10; B4a, B4b, B4c, B4d
Simple primary galvanic cells: electrodes, electron and ion flow, salt bridge	A9; B2d, B2f, B3a–e
Linked skills: redox stoichiometry (mass ↔ mol)	A8; B2e